METRIC
MIL-PRF-49446B(CR)
3 JULY 1997
SUPERSEDING
MIL-D-49446A(CR)
6 November 1987

# PERFORMANCE SPECIFICATION
# DETECTOR-DEWAR, DT-591A/UA

This performance specification is approved for use by USACECOM, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This performance specification covers one type of common module, Detector-Dewar, DT-591A/UA, herein referred to as the Detector-Dewar.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Handbooks. The following handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2). Handbooks are for guidance only and therefore are not mandatory.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications- Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-454 - Standard General Requirements for Electronic Equipment.

(Unless otherwise indicated, copies of the above specification, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following Government documents, drawings, and publications form a part of this specification to the extent specified herein. Thedrawing listed below is identified to ensure accurate interface of the item produced under this performance specification with existing hardware.

PURCHASE DESCRIPTION

DEPARTMENT OF DEFENSE

B2-2303020104A -Cooler, Cryogenic, Mechanical Stirling, (HD-1132/UA)

DRAWINGS

USA CECOM

54490-5004385 -Detector-Dewar DT-591A/UA

(Copies of documents, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of documents not which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

"Standard Procedure for Testing infrared Detectors and Their Performance", R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering.

(Copies of the above non-government publication should be obtained from the procuring activity or as directed by the contracting officer.)



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AMERICAN SOCIETY FOR QUALITY CONTROL (ANSI/ASQC)

ANSI/ASQC Z 1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Application for copies should be addressed to the American National Standards Institute 1430 Broadway, New York, NY, 10018.)

2.4 Order of precedence. In event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. Where specified in the contract (see 6.2), the contractor shall furnish first article units for inspection (see 4.2).

3.2 Materials. The contractor shall select materials that are fully capable of meeting all of the operational and environmental requirements specified herein. The materials specified in the applicable drawings are recommended, but are not mandatory. Selection criteria for the class, grade or type part shall ensure that the material will be able to perform its intended function when it is assembled. Verification of the supplier meeting the overall performance requirements shall be the governing acceptance standard. Recovered materials shall be used to the maximum extent possible.

3.3 Description. The Detector-Dewar is an opto-electronic device sensitive to radiation in the 7.5 to 12.0 micrometer spectral region. The detector bias resistors shall be an integral part of the Detector-Dewar.

3.3.1 Design. The Detector-Dewar shall meet form, fit, function, and interface requirements of this specification.

3.3.2 Weight. Maximum weight of the Detector-Dewar shall be 227 grams (0.50 Lb.), including bias resistors.

3.3.3 Dimensions. Dimensions and dimensional tolerances of the Detector-Dewar shall be in accordance with Drawing 5004385 to ensure accurate physical interface of the item produced under this specification with the existing hardware.

3.4 Components. The Detector array shall rotate 360° with respect to the Dewar mounting flange. Two or more reusable, electrically activated getters shall be an integral part of the Detector-Dewar for maintenance of the vacuum. Each getter shall withstand at least seven (7) firings in operational use. Activation current (amperes) and time (minutes) shall be provided



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on the label as shown in View K of 5004385. Data shall be marked in contrasting characters on a background in accordance with Note 27 of 5004385. The Detector-Dewar shall consist of a 60-element detector array and a resistor bias pack.

3.5 Performance characteristics.

3.5.1 Detector-Dewar defects. Detector-Dewar shall operate 80°, +5/ -20° K. The Detector-Dewar shall contain no more than two (2) defective elements. Elements 85-96 shall be defect free. Defects shall be separated from one another by at least four (4) nondefective elements. Elements that are open (dead) or have a D*BB of less than $5.0 \times 10^9$ shall be electrically strapped to an adjacent element with common ground. Perform 4.5.1 for acceptance.

3.5.2 Time constant. The time constant is the time for the element response to reach 63% of its steady state value when irradiated by a 7.5-12 micrometer source producing a signal to noise ratio of <100:1. The time constant of each detector element shall be less than two (2) microseconds. Perform 4.5.2 for acceptance.

3.5.3 Sensitive area. The physical area shall be 0.0960 by 0.07440 inch or greater as determined by an optical microscope or similar method. Perform 4.5.3 for acceptance.

3.5.4 Crosstalk. The Crosstalk signal level between any two (2) elements shall be less than -26 dB as determined by measuring 4 elements. Perform 4.5.4 for acceptance.

3.5.5 Blackbody detectivity. The Blackbody detectivity of all nondefective elements shall be at least $7.5 \times 10^9$ an $Hz^{1/2}$ $w^{-1}$ except for channels 85-96 which shall not be less than $9.0 \times 10^9$ cm $Hz^{1/2}$ $w^{-1}$ when the Dewar is cooled with $LN_2$ to 77 ± 3° K. Perform 4.5.5 for acceptance.

3.5.6 Noise. The detector noise of each element shall be at least $2.5 \times 10^{-9}$ volts RMS $Hz^{1/2}$. Perform 4.5.6 for acceptance.

3.5.7 Responsivity. The average 500° K blackbody responsivities of the acceptable elements shall not be less than $8.3 \times 10^3$ volts/watt, nor greater than $20 \times 10^3$ volts/watt, when the Dewar is cooled with $LN_2$ to 77°, ±3° K. The blackbody responsivity of each element shall not vary by more than ±20 percent of the mean value of the elements which are not defective (see 6.4.2). The nominal detector element size shall be used in responsivity calculations and blackbody temperature shall be 500°, ±5° K (see 6.3.7). Perform 4.5.7 for acceptance.

3.5.8 Temperature sensors. Temperature sensors shall be installed at or near the detector array location in the Dewar to provide an electrical signal which is an analog of temperature. The temperature sensor shall have a junction voltage of 1.060, ±0.015 volts with a 1.00 mA, ±0.01 mA forward current when the Detector-Dewar is operated at 77°, ±3° K. The voltage shall be recorded on the appropriate label on the Detector. Perform 4.5.8 for acceptance.



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3.5.9 Spectral response. Each element of the Detector-Dewar shall have special sensitivity, defined as the 50 percent relative response points, of 7.7, ±0.3 to 11.7, ±0.5 micrometers. Perform 4.5.9 for acceptance.

3.5.10 Detector bias. The average bias current of the array shall not exceed 3.0 milliamperes per element, and no one element shall take a bias current of more than 4.0 milliamperes. The total bias power dissipated by the array shall not exceed 30 milliwatts. The detector element shall be biased with current flow in the same direction. The detector bias pack shall be operated at + 3.0V+/0,2Vdc. Perform 4.5.10 for acceptance.

3.5.11 Dewar warm and cold shields. The incorporation of an internal cold shield is optional. If incorporated, the Detector-Dewar shall meet the following Field Of View (FOV) requirements: The minimum half angle FOV from the center of the end elements of the array in the elevating direction shall be 14 degrees. In the direction perpendicular to the array's axis, the cold shield(s) shall allow for each element, no vignetting of any ray to any point on the detector surface within an angle of 17 degrees. This shall be measured from and on either side of a line perpendicular to the array's axis and parallel to the optical axis. Perform 4.5.11 for acceptance.

3.5.12 Heat load. The detector heat load shall not exceed 120 milliwatts when all detector elements are operating at their final bias current and the Dewar is cooled to $80^{o}$, ±5°K.

3.5.13 Vacuum integrity. The Dewar shall comply with heat load requirements of 3.5.12 for a period of at least 0ne (1) year. Perform 4.5.13 for acceptance.

3.5.14 Microphonics. When cooled to a temperature of 60-80°K, with full operating bias applied in accordance with paragraph 3.5.10, and vibrated in accordance with paragraph 3.5.14.1, all elements of the Detector-Dewar shall have no noise peaks above the white noise reference value greater than 8dB from 300 to 10,000 Hz, and 12dB from 10,001 to 20,000 Hz. Perform 4.5.14 and 4.5.14.1 for acceptance.

3.5.14.1 Microphonics vibration. The Detector-Dewar shall be coupled to a mechanical cryogenic refrigerator that produces random vibrational excitation of the Detector-Dewar in three mutually perpendicular axes. The frequency distribution of the vibrations shall be in a range from 0 to 20,000 Hz. The RMS sum of the three axis vibration shall be at least of 1.0 g rms. Each axis shall produce a minimum vibrational excitation of 0.4 g rms. In the frequency range of 2,000 to 20,000 Hz, the RMS sum shall be at least 0.8 g rms. Perform 4.5.14.1 for acceptance.

3.5.15 Apparent thermal mass (cooldown). When mated to a cryogenic cooler capable of cooling a test mass of 1.25 grams (250 joules) of aluminium (Figure 3) from ambient (296°K) to 80°K in not less than 5, nor greater than 6 minutes, the Dewar cooldown time from ambient temperature to 80°K shall not exceed the time achieved with the test mass.

3.6 Environmental conditions.



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3.6.1 Temperature shock. The Detector-Dewar shall not be damaged (see 6.3.6) by sudden changes in temperature between -54° C and +71° C.(see 6.2.e (1) )

3.6.2 High temperature. The Detector-Dewar shall not be damaged (see 6.3.6) by storage or operation to 71°C (see 4.6.2).

3.6.3 Low temperature. The Detector-Dewar shall not be damaged by operation to -54° C or storage to -62°C (see 4.6.3).

3.6.4 Shock. The Detector-Dewar shall not be damaged by high intensity shocks of 100 g's peak amplitude with 11 milliseconds duration and by bench handling tests (see 4.6.4).

3.6.5 Vibration. The Detector-Dewar shall not be damaged by vibrations over the frequency spectrum at the specified g levels and amplitudes shown in Figure 1 (see 4.6.5).

3.7 Reliability. The Detector-Dewar shall have a specified mean time between failure (MTBF) of 27,600 hours..

3.8 Treatment and painting. Unless otherwise specified, the Detector-Dewar shall have an external surface of a composition or surface treatment which will prevent rust, corrosion, and fungus growth in the full range of environmental requirements. External color and surface shine shall reduce light reflection and reduce recognition. Surface treatment dherence must be assured through the full range of environmental and use factors.

3.9 Preconditioning. Each Detector-Dewar shall be subjected to vibration and temperature cycling prior to Group A testing as specified in Figure 2.

3.10 Nameplates and product marking. The Detector-Dewar module, parts, components, subassemblies, and assemblies thereof shall be marked for identification and reference designation markings in accordance with View J of 5004385. Marking shall be positioned as shown in 5004385.

3.11 Workmanship. All components and parts shall be manufactured and assembled to assure uniform quality and be free of defects. A recommended criteria for workmanship is contained in the applicable portion of the equipment drawings and the information in MIL-HDBK-454. See also 6.3.6.

4. VERIFICATION

4.1 Classification of inspections. The inspections shall be classified as follows:

a. First article inspection (4.2).

b. Conformance inspection (4.3).



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4.2 First article inspection. When specified in the contract, first article inspection shall be performed by the contractor on a minimum of five (5) preconditioned (see 3.9) sample Detectors-Dewar. This inspection shall include the visual examination of 4.2.1 and the tests of 4.2.2 through 4.3. Refer to the contract for specific instructions regarding first article inspection (see 6.2).

4.2.1 Visual examination. All completed assemblies shall be examined for conformance to the applicable specifications or drawings (See Table I)..

TABLE I. Inspections.

| Inspection | Requirement |
|---|---|
| Dimensions not as specified | 3.3.1 |
| Any as specified part or component missing or damaged | 3.3.1 |
| Weight not specified | 3.3.2 |
| Getters and label not as specified | 3.4 |
| Detector orientation range not specified | 3.4 |
| Sensitive area not as specified | 3.5.3 |
| Dewar warm and cold shields not as specified | 3.5.11 |
| Treatment and painting not as specified | 3.8 |
| Marking not as specified | 3.10 |
| Workmanship not as specified | 3.11 |

4.2.2 Tests. First article inspection shall be conducted upon units which have successfully completed the inspections specified in 4.2.1. Tests may be conducted in any order, except that microphonics shall be performed after the shock and vibration test (see 4.6.4 and 4.6.5).



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TABLE II. First article inspection.

| Inspection | Requirement paragraph | Verification paragraph |
|---|---|---|
| Detector-Dewar defects | 3.5.1 | 4.5.1 |
| Time constant | 3.5.2 | 4.5.2 |
| Crosstalk | 3.5.4 | 4.5.4 |
| Detectivity | 3.5.5 | 4.5.5 |
| Detector resistance | 3.5.6 | 4.5.6 |
| Responsivity | 3.5.7 | 4.5.7 |
| Temperature response | 3.5.8 | 4.5.8 |
| Spectral response | 3.5.9 | 4.5.9 |
| Detector bias | 3.5.10 | 4.5.10 |
| Heat load | 3.5.12 | 4.5.12 |
| Vacuum integrity | 3.5.13 | 4.5.13 |
| Microphonics | 3.5.14 | 4.5.14 |
| Apparent thermal mass (Cooldown) | 3.5.15 | 4.5.15 |
| Temperature shock | 3.6.1 | 4.6.1 |
| High temperature | 3.6.2 | 4.6.2 |
| Low temperature | 3.6.3 | 4.6.3 |
| Shock | 3.6.4 | 4.6.4 |
| Vibration | 3.6.5 | 4.6.5 |
| Reliability | 3.7 | 4.7 |

4.2.3 Disposition of first article samples. First article sample shall not be considered as part of the procurement quantities (see 6.2).

4.2.4 Inspection procedures. Unless otherwise specified, the Detector-Dewar shall be inspected at ambient temperature (see 6.3.8) when operated at temperature of 80°K, ±5°K, and biased to +3.0 Vdc, ±0.2 Vdc.

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.5 and the Group inspections of 4.5.2 et el.

4.4 Examination. Each Detector-Dewar shall be examined for compliance with the requirements specified in section 3. et el. Any redesign or modification following failure to meet the specified requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any special requirements or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4.1 Visual examination. Examination shall be as specified in 4.2.1.



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4.4.2 Group inspections.

4.4.2.1 Group A inspection. Group A inspection shall be conducted on all (100 percent) of the Detector-Dewars following successful inspection of 4.5.1 and preconditioning of 3.9. Group A tests, listed in Table III, may be performed in any order except that Detector-Dewar defects (3.5.1) shall be run last.

TABLE III Group A inspection.

| Inspection | Requirement paragraph | Verification paragraph |
|---|---|---|
| Detector-Dewar defects | 3.5.1 | 4.5.1 |
| Detectivity | 3.5.5 | 4.5.5 |
| Responsivity | 3.5.6 | 4.5.6 |
| Detector resistance | 3.5.7 | 4.5.7 |
| Temperature sensors | 3.5.8 | 4.5.8 |
| Detector bias | 3.5.10 | 4.5.10 |
| Heat load | 3.5.12 | 4.5.12 |
| Microphonics | 3.5.14 | 4.5.14 |

4.4.2.2 Group B inspection. Not required.

4.4.2.3 Group C inspection. Group C inspection shall be conducted on Detector-Dewars selected from units which have passed the inspection in 4.4.2.1. The measurement and verification of performance requirements 3.5.9, 3.5.13, 3.5.15 and 3.6.5 shall be performed on a sample quantity (4.5.23.1) through the test requirements 4.5.9, 4.5.13, 4.5.15, and 4.6.5.

4.5.2.3.1 Sampling of group C inspection. The tests of 4.5.9, 4.5.13, 4.5.15 and 4.6.5 shall require 100 percent inspection of the entire lot for the failed attributes only, with continued follow-on lots inspected until zero failures occur in any lot. Upon achieving zero failure at 100 percent inspection, follow-on lot inspection shall be conducted at 10 percent sampling. Thereafter, upon achievement of zero failure during 10 percent samples, the sample rate shall be lowered to 3 percent of the specified lot or one unit, whichever is larger. If one or more failures occur during the 10 percent inspection sampling, the 100 percent inspection testing shall be reinstituted.

4.4.2.3.2 Group C failures. Action required relative to group C failures shall be as specified the contract or purchase order (see 6.2.).

4.4.2.3.3 Disposition of group C samples. Group C samples shall be accepted on contract subsequent to successful completion of testing of Table III.

4.4.2.4 Group D inspection. Not required.



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4.5 Test methods.

4.5.1 Detector-Dewar defects. The detector array shall be checked for defects. Failure to meet requirements of 3.5.1 shall constitute failure of this test.

4.5.2 Time constant. Measurement shall be performed on two (2) elements of the array each separated by at least 20 elements. This time constant is defined as the time required for the element response to reach 63 percent of its steady state value when irradiated by a 7.5 to 12 micrometer source which produces a signal-to-noise ratio not greater than 100:1. This time constant shall be considered equal to the time constant defined by the high frequency rolloff of the noise spectrum of the detector. Any measurement technique consistent with these definitions shall be considered acceptable. Failure to meet requirements of 3.5.2 shall constitute failure of this test.

4.5.3 Sensitive area. The physical area of 0.0960 by 0.0144 inch or greater shall be determined by optical microscope examination or equivalent measurement method. Measurement shall be made on two (2) elements of the array each separated by at least 20 elements prior to the integration of the Dewar window with the Detector-Dewar. Failure to meet requirements of 3.5.3 shall constitute failure of this test.

4.5.4 Crosstalk. Measurements shall be made on four (4) elements of the array. The crosstalk shall be measured under the following conditions:

a. An element shall be irradiated by a 7.5 to 12 micrometer source having a signal 20 times greater than the noise level for the element. The image of the source shall fall entirely within the area of the detector element.

b. The signal level of the element adjacent to and on the same ground path of the irradiatedelement shall be measured. The element under test shall meet the requirements of 3.5.4. Failure to meet requirements of 3.5.4 shall constitute failure of this test.

4.5.5 Blackbody detectivity. Noise measurements shall be made on each element of the array under the following conditions: The detector cooled to 77 ±3 K with $LN_2$, 295, ±5 K background temperature and equivalent noise bandwidth of 9 kHz ±200 Hz with low frequency 3dB point (high pass) no greater than 80 Hz. Using the results of these measurements, the results of the 500 K blackbody responsivity measurements, and the nominal detector size (see 6.3.7), the 500 K blackbody detectivity for each element of the array shall be calculated. Failure to meet requirements of 3.5.5 shall constitute failure of this test.

4.5.6 Detector resistance. The resistance of each element of the array shall be measured. Failure to meet requirements of 3.5.6 shall constitute failure of this test.

4.5.7 Responsivity. Blackbody responsivity of each element of the array may be measured with the Dewar cooled with $LN_2$ to 77°K, ± 3° K. Nominal element size (see 6.3.7)



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shall be used in responsivity calculations and blackbody temperature shall be 500°, ±5°K. Failure to meet requirements of 3.5.7 shall constitute failure of this test.

4.5.8 Temperature sensors. The 77° K calibration point may be determined by cooling the inner Dewar with liquid nitrogen. A 1.00 mA ±0.01 mA bias shall be applied to the sensor diode and the voltage shall be monitored and recorded at the 77° K calibration point. Failure to meet requirements of 3.5.8 shall constitute failure of this test.

4.5.9 Spectral response. The spectral response of the sixth element from each end of the array or adjacent element on the high end and the higher number of the two center elements shall be tested per "Standard Procedures for Testing Infrared Detectors and Describing Their Performance", by R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. If these elements are defective an adjacent element may be substituted. The spectral sensitivity is defined as the 50 percent relative response points and shall be measured on the assembled Detector-Dewar. Failure to meet requirements of 3.5.9 shall constitute failure of this test.

4.5.10 Detector bias. The detector bias power shall be tested per "Standard Procedures for Testing Infrared Detectors and Describing Their Performance", by R.C. Jones, D. Goodwin, and G. Pullen, dated 12 September 1960, Office of the Director of Defense Research and Engineering. Failure to meet requirements of 3.5.10 shall constitute failure of this test.

4.5.11 Dewar warm and cold shields. During in-process inspection, the warm and cold shield shall be measured to ensure conformance with the requirement Alignment of the cold shield relative to detector elements shall be checked 100 percent during in-process inspection. Failure to meet requirements of 3.5.11 shall constitute failure of this test.

4.5.12 Heat load. The heat load of the Detector-Dewar shall be determined with all detectors operating at specified bias. The heat load of the Detector-Dewar shall be determined by partially filling the Dewar's inner stem with liquid nitrogen and monitoring the rate of loss of the cryogen. The heat load shall be calculated from the average or integrated loss rate of the cryogen for the last 5 minutes of the Dewar's hold-time. For calculations, cooling produced by the nitrogen phase change only shall be considered as total useful energy. If this test is performed with the detector array unbiased, the results of this test shall be added to the results of the detector bias power test to determine the heat load. Failure to meet requirements of 3.5.12 shall constitute failure of this test.

4.5.13 Vacuum integrity. Vacuum integrity shall be determined on a basis of heat load capacity. Heat load shall be measured on the unbiased Detector-Dewar. The temperature of the Detector-Dewar shall then be elevated to +71°C for a period of 72 hours. At the completion of the temperature exposure and stabilization at ambient temperature, the heat load shall be measured again. Getters may not be fired during this test. Failure of the Detector-Dewar to meet the requirements of 3.5.12 (heat load) or a 15 percent, or greater, increase in heat load between the measurement before and after temperature exposure shall constitute failure of this test.



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4.5.14 Microphonics. Microphonics shall be measured while the Detector is vibrated in accordance with paragraph 4.5.14.1. The Detector-Dewar shall be cooled to 60-80$^{o}$K with full operating bias applied. Mounting to the cooler shall be made by tightening the mounting screws to a torque of 3.50 inch pounds minimum. Vibration excitation shall be measured by an accelerometer mounted on the mounting plate of the Detector-Dewar. The white noise reference level may be determined by averaging the noise in each channel from 12,000 to 20,000 Hz with vibration applied. Noise spectra shall be measured using a bandwidth between 50 Hz and 365 Hz. Failure of any channel to meet the requirements of 3.5.14 shall constitute a defect as defined in paragraph 3.5.1.

4.5.14.1 Microphonics Vibration. The Detector-Dewar shall be coupled to a mechanical cryogenic refrigerator that produces random vibrational excitation of the Detector-Dewar in three mutually perpendicular axis. The frequency distribution of the vibrations shall be in a range from 0 to 20,000 Hz. The RMS sum of the three axis vibration shall be at least of 1.0g RMS. Each axis shall produce a minimum vibrational excitation of 0.4 g RMS. In the frequency range of 2,000 to 20,000 Hz, the RMS sum shall be at least 0.8 g RMS. The mechanical cryogenic refrigerator may be mated to the Dewar with a bellows.

4.5.15 Apparent thermal mass. (Cooldown). An HD-1132/UA cryogenic cooler shall be mated to a standard test mass (Figure 3). The capacity of the HD-1132/UA shall be such that the 1.25 gram (250 joules) test mass is cooled from ambient (296$^{o}$K) temperature to 80$^{o}$K ±1.5$^{o}$ K in not less than 5, nor greater than 6, minutes. This measurement shall be made using a highly polished metal Dewar without any radiation shielding. This test Dewar and test mass shall consist of the following: A high vacuum of less than 10 microtorr, and a 1.25 gram aluminum test mass. The time to cool the thermal mass shall be recorded. The voltage output of the temperature sensor diodes in the Detector-Dewar under test shall be measured at 77$^{o}$ ± 1$^{o}$K and at ambient temperature. The voltage output at 80$^{o}$K shall then be interplated. The cooler shall then be coupled to the Detector using thermal grease and bellows. Operating bias power shall be applied and the time for the temperature of the diode to reach 80$^{o}$K shall be recorded. Failure of the diode to cool within the requirements of 3.5.15 shall constitute failure of this test.

4.6 Environmental tests. Environmental testing methods contained in 6.2.d are acceptable methods for determining compliance with Section 3.6 et el. The contractor may elect to use alternative methods and testing procedures. In such cases, the contractor shall be required to show where his own tests and measurements for verifying the environmental conditions are equivalent to the methods cited herein. When the contractor desires to deviate from these cited methods, he shall be prepared to show equivalency via a test plan which has been approved by the Government contracting office. Also, environmental tests shall be accomplished using known testing apparatus such as an environmental test chamber, vibration test table, pendulum shock device, or similar test equipment for which the technical capabilities to properly simulate extreme climatic or transportation conditions are scientifically documented. The methods for testing to verify compliance with all stated performance characteristics selected by the contractor shall be equivalent to the recommended procedures found herein.



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4.6.1 Temperature shock. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.1. A recommended method for verification may be found in 6.2.

4.6.2 High temperature. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.2. A recommended method for verification may be found in 6.2.

4.6.3 Low temperature. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.3. A recommended method for verification may be found in 6.2.

4.6.4 Shock. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.4. A recommended method for verification may be found in 6.2.

4.6.5 Vibration. Verify that the Detector-Dewar meets the requirements of paragraph 3.6.5. A recommended method for verification may be found in 6.2.

4.7 Reliability. A reliability prediction and math model shall be maintained by the contractor which reflect the current design and components of the Detector-Dewar. Failure of the prediction or model to explicitly delineate compliance with this section shall constitute failure of the reliability requirement.

## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department system Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The Detector-Dewar is intended for use in infrared systems.

6.2 Acquisition requirements. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. First article tests are required except for existing suppliers (see 4.2.2 ).

(1) Time frame for submission of first article test reports when first article tests are



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required.

(2) Time frame for approval of first article test reports when first article tests are required.

(3) First article reliability testing, if required.

c. Production delivery schedule.

(1) Defined in terms of monthly lots.

(2) Actions required relative to Group C failures (see 4.4.2.3.2).

d. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.2.2 and 2.3).

e. Contractor may use MIL-STD-810 for environmental testing. Apply as follows:

(1) Measurement may be in accordance with commercial environmental test methods equal or similar to MIL-STD-810, method 503.1, Procedure I, except that the temperatures of step 2 should be -54°C and the temperature of steps 1 and 4 should be +71°C. (see 4.7)

(2) Measurement may be in accordance with commercial environmental test methods equal or similar to MIL-STD-810, method 501.1, Procedure I except that the temperature of steps 4 and 5 should be +71°C. (see 4.7)

(3) Measurement may be in accordance with commercial environmental test methods equal or similar to MIL-STD-810, method 502.1, Procedure I except that the storage temperature of step 2 should be -62°C and the low operating temperatures of step 4 should be -54°C. (see 4.7)

(4) Measurement may be in accordance with commercial environmental test methods equal or similar to MIL-STD-810, method 516.2, Procedure IV, Figure 516.2-1 (100 g's peak value at 11 milliseconds time duration ) and Procedure V. (see 4.7)

(5) Measurement may be in accordance with commercial environmental test methods equal or similar to MIL-STD-810, method 514.2, Procedures I and VIII except that the curve shown in Figure 1 of this specification should be used in lieu of the MIL-STD-810 curves. The sinusoidal cycle time should be 120 minutes/axis. The dwell time should be 1/6 of the cycling time at each axis.

f. Detector-Dewar drawing numbers.

g. When First Article rough handling tests are not required.



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h. Deliver all first article samples to the procuring activity.

i. Group C inspection monthly sample size to be established based on cumulative production totals of concurrent contracts.

j. When First Article Packaging Inspection reports require acquisition activity approved prior to production unit packing.

k. Sampling plans for Groups B and C inspections.

6.3 Definitions.

6.3.1 Element. An element is defined as one channel in the 60-element Detector-Dewar.

6.3.2 Defective element. A defective element is one that fails to meet any one of the requirements of 3.5.5, 3.5.5.1, 3.5.6, 3.5.7, and 3.5.14.

6.3.3 Strapping. Strapping is defined as electrically connecting a defective element to a non-defective element. The pair of strapped elements has a 100 percent Crosstalk and is considered as a single defect. Element whose D* is less than $5.0 \times 10^{9}$ cm $Hz^{1/2}$ $w^{-1}$ shall be opened prior to strapping.

6.3.4 Array axis. The array is defined as a line on the surface of the array passing through the center of the projected geometrical areas of each element.

6.3.5 Optical axis. The Detector-Dewar's optical axis is defined as a line perpendicular to the detector mounting plate that intersects the array axis at the midpoint of the array.

NOTE: One-half of the active detector elements are located on each side of the optical axis.

6.3.6 Damage. Breakage, loosening, shifting, evidence of corrosion, or failure of any finish, hardware, connection or component; and any degradation of Detector-Dewar performance to values less than specified herein.

6.3.7 Nominal detector element size. Defined as 0.0016 by 0.0023 inch.

6.3.8 Ambient temperature. Ambient temperature is considered to be +23° C, ±10° C.

6.3.9 Specified lot quantity. The specified lot quantity will be one-month's production unless the contract specifies a different lot basis such as one-week's production or those units processed as single batch.



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6.3.10 Variation of responsivity with temperature. When coupled to a cryogenic refrigerator, the temperature of the array may reach 55° K. Under those conditions, the Responsivity (paragraph 3.5.7) may decrease to as low as 5.6 x $10^3$ volts/watt.

6.3.11 Microphonic test caution. The microphonic requirements and tests specified herein are intended to be representative of an operational environment and combat scenario for detectors mounted in a well designed system using current technology. If the system vibration levels are increased above those specified, the microphonic noise may increase.

6.3.12 Detector temperature. Detector temperature is defined as that indicated by the temperature sensor diode.

6.4 Subject term (keyboard listing).

Flir Systems
Opto-electronic
Sensors

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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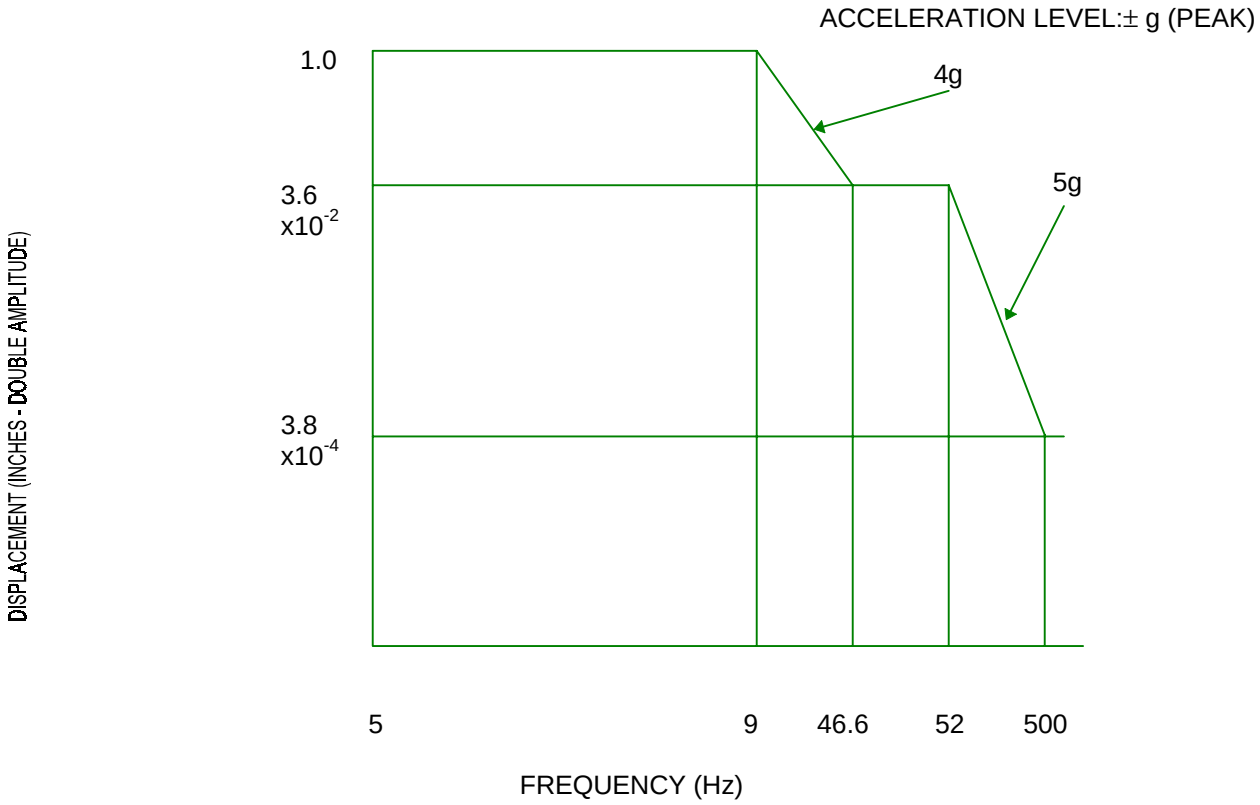
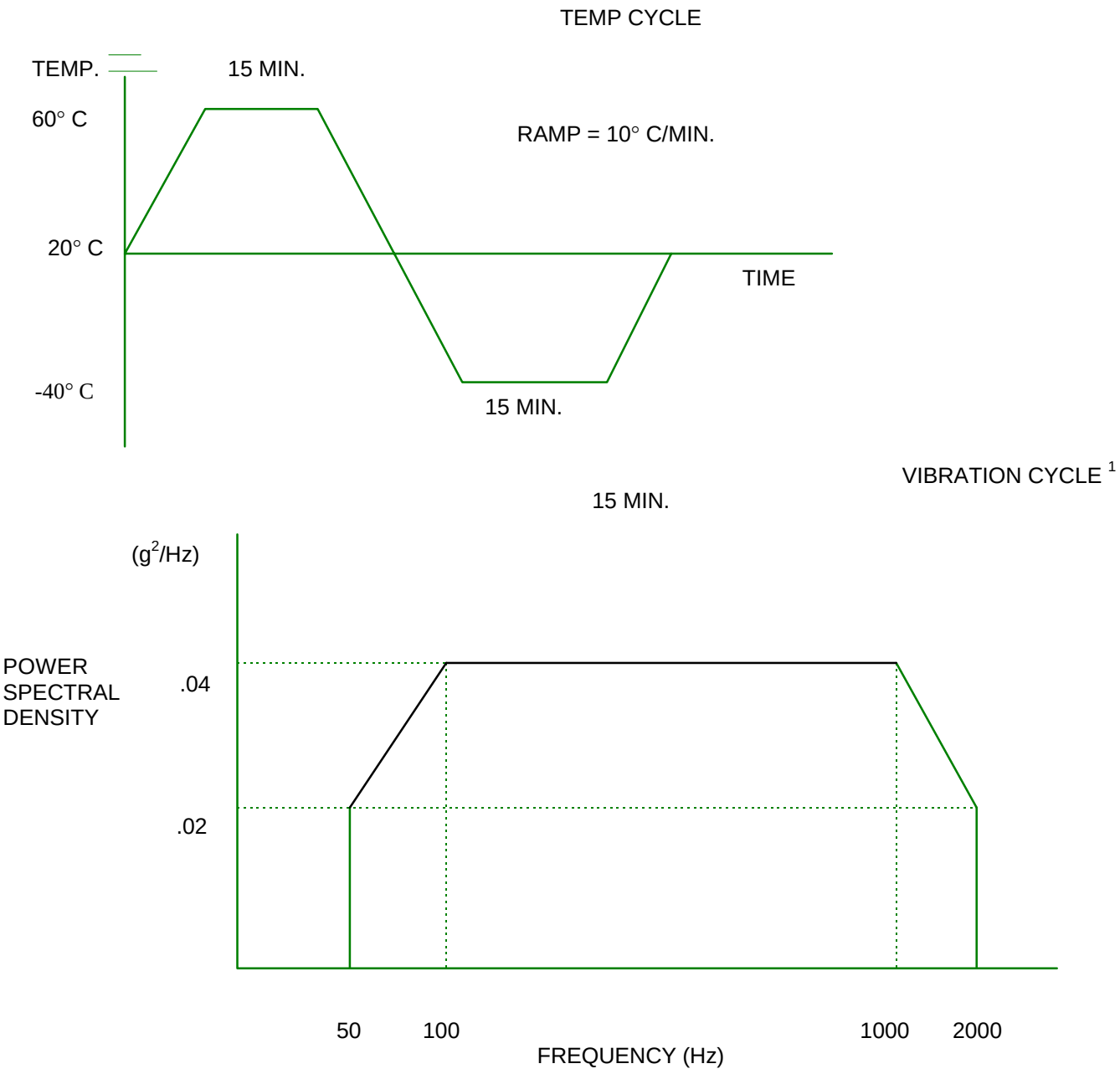



FIGURE 1. Vibration Inspection Profile



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Notes:

1. Vibration performed along detector cold well axis.
2. Temp. cycle precedes vibration cycle.

FIGURE 2. Preconditioning Cycle



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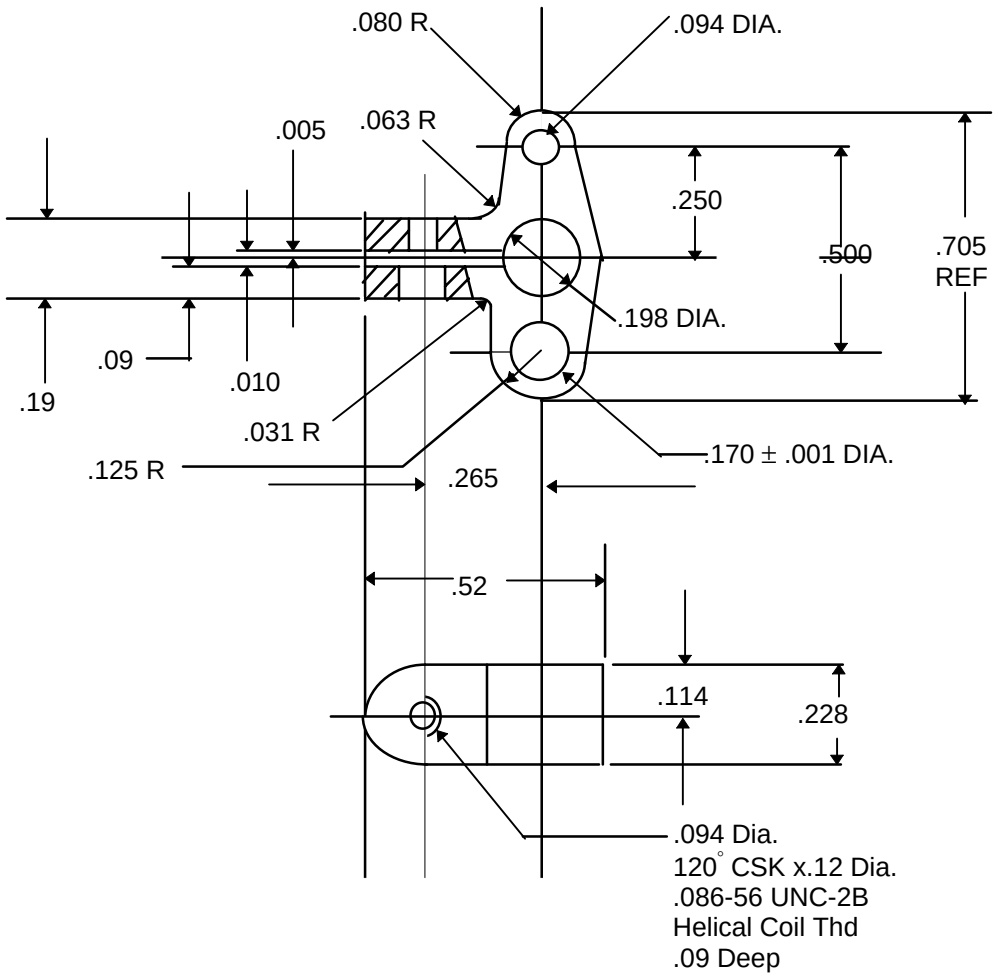



FIGURE 3. Aluminum Block (250 Joule Mass)



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**STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL**

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER** MIL-PRF-49446B | **2. DOCUMENT DATE (YYMMDD)** 97/07/03 |
|---|---|---|

**3. DOCUMENT TITLE** DETECTOR-DEWAR, DT-591A/UA

**4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)**

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

**8. PREPARING ACTIVITY**

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|---|---|
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